

# Math vs. Your Eyes

 **Green Level: I Want More Support**

Name: \_\_\_\_\_ Date: \_\_\_\_\_ Period: \_\_\_\_\_

## Can Math Prove Your Eyes Wrong?

Your eyes can be tricked by optical illusions—but a ruler never lies! Today you'll use math to prove that what you SEE isn't always what's REAL.

### Activity 1: The Müller-Lyer Illusion

Look at these two lines. Which one looks LONGER?



**My prediction:** (circle one)

Top line is longer   |   Bottom line is longer   |   Same length

**Now measure with your ruler!**

Top line: \_\_\_\_\_ cm   Bottom line: \_\_\_\_\_ cm

**Was your prediction correct?** Yes / No

**Why does this illusion work?** The arrows create *angles* that trick your brain into thinking one line is closer (and shorter) than the other!

### Activity 2: Simple Perspective Drawing

Follow these steps to draw a 3D box using math:

1. Put a dot in the center of the box below (this is your **vanishing point**)
2. Draw a square in the bottom-left corner
3. Draw lines from each corner of the square to the vanishing point
4. Draw a smaller square where those lines would end

You just created the illusion of depth using geometry!

*Draw your perspective box here*

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## What Did You Learn?

Fill in the blanks:

The two lines in the Müller-Lyer illusion are actually \_\_\_\_\_ (the same / different).

The arrows create \_\_\_\_\_ that trick your brain.

When I used a ruler, I found out that math is more \_\_\_\_\_ than my eyes.

## Why Does Math Matter?

Think of times when measuring is more reliable than just looking:

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## Mini Museum Connection

The stereoscope in the Mini Museum uses math too! Your eyes are about 6 cm apart, so each eye sees a slightly different angle. The stereoscope shows each eye a different picture, and your brain calculates depth from the difference!

Visit: [minimuseumproject.com/exhibitions/seeing-is-deceiving/illusions](https://minimuseumproject.com/exhibitions/seeing-is-deceiving/illusions)

## Try This Math Problem!

The stereoscope cards were photographed with cameras 6.5 cm apart (the average distance between human eyes).

If YOUR eyes are 6 cm apart, what is the difference? \_\_\_\_\_ cm

What percentage is that? (Hint: divide the difference by 6.5, then multiply by 100)

Answer: \_\_\_\_\_ %

## Share with Your Family!

Show your family the Müller-Lyer illusion. Ask them which line is longer, then measure to prove the truth! See whose eyes are tricked the most.